

Topic B13

Coordination and response

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B14.1	Nervous control
B14.2	Sense organs
B14.3	Hormones
B14.4	Homeostasis
B14.5	Plant responses

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Nervous system

Receptors detect stimuli; impulses travel along sensory neurones to the central nervous system, across synapses using neurotransmitters, then along motor neurones to effectors. Reflexes are rapid automatic responses that protect the body.

Examiner insight

Give the complete reflex arc in the correct order.

2. Eye

The cornea and lens refract light onto the retina; the iris controls pupil size; rods detect low light and cones detect colour. For near objects, ciliary muscles contract, suspensory ligaments slacken and the lens becomes thicker and more refractive.

Examiner insight

Accommodation changes lens shape, not the distance between lens and retina.

3. Hormones and homeostasis

Hormones are chemical messengers transported in blood. Insulin lowers blood glucose by increasing uptake and glycogen formation; glucagon raises it by promoting glycogen breakdown. Negative feedback reverses deviations from a set range. Adrenaline prepares the body for action.

Examiner insight

State stimulus, coordinator, effector and corrective response in homeostasis.

4. Plant responses

Shoots are positively phototropic and roots positively gravitropic. Auxin made in shoot tips moves to the shaded side and stimulates greater cell elongation, bending the shoot toward light. In roots, high auxin inhibits elongation, contributing to downward curvature.

Examiner insight

The same auxin distribution can have different effects in shoots and roots.

Worked application

Situation: Blood glucose rises after a meal.

Prediction: Insulin secretion increases and glucose falls toward normal.

Reason: Cells take up glucose and liver or muscle convert it to glycogen by negative feedback.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B13

Coordination and response

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines stimulus? [1]
 - A a change in the internal or external environment detected by a receptor
 - B a cell or sense organ detecting a stimulus
 - C a muscle or gland that carries out a response
 - D maintenance of a stable internal environment
2. Which statement best defines receptor? [1]
 - A a muscle or gland that carries out a response
 - B a cell or sense organ detecting a stimulus
 - C maintenance of a stable internal environment
 - D a rapid automatic response to a stimulus
3. Which statement best defines effector? [1]
 - A maintenance of a stable internal environment
 - B a rapid automatic response to a stimulus
 - C a muscle or gland that carries out a response
 - D a change in the internal or external environment detected by a receptor
4. Which statement best defines homeostasis? [1]
 - A a rapid automatic response to a stimulus
 - B a change in the internal or external environment detected by a receptor
 - C a cell or sense organ detecting a stimulus
 - D maintenance of a stable internal environment
5. Which statement best defines reflex? [1]
 - A a rapid automatic response to a stimulus
 - B a change in the internal or external environment detected by a receptor
 - C a cell or sense organ detecting a stimulus
 - D a muscle or gland that carries out a response
6. Which statement about nervous system is correct? [1]
 - A A muscle carrying out a response.
 - B Receptors detect stimuli; impulses travel along sensory neurones to the central nervous system, across synapses using neurotransmitters, then along motor neurones to effectors.
 - C A gland producing every hormone.
 - D A sensory neurone.

7. Which statement about eye is correct? [1]
- A A gland producing every hormone.
 - B A sensory neurone.
 - C The cornea and lens refract light onto the retina; the iris controls pupil size; rods detect low light and cones detect colour.
 - D Keeping every condition completely constant.
8. Which statement about hormones and homeostasis is correct? [1]
- A A sensory neurone.
 - B Keeping every condition completely constant.
 - C A learned voluntary action only.
 - D Hormones are chemical messengers transported in blood.
9. Which statement about plant responses is correct? [1]
- A Shoots are positively phototropic and roots positively gravitropic.
 - B Keeping every condition completely constant.
 - C A learned voluntary action only.
 - D A muscle carrying out a response.
10. Which prediction is most appropriate when blood glucose rises after a meal? [1]
- A A muscle carrying out a response.
 - B A sensory neurone.
 - C Insulin secretion increases and glucose falls toward normal.
 - D A learned voluntary action only.

Section B · Structured questions

11 Biologists use precise vocabulary when studying coordination and response.

(a) Define stimulus.

[2]

(b) Define receptor.

[2]

(c) Distinguish effector from the incorrect idea: “a sensory neurone”.

[2]

(d) State two important features of nervous system.

[2]

[Total: 8]

12 Blood glucose rises after a meal.

(a) State the most likely biological outcome.

[2]

(b) Explain the outcome using a linked biological mechanism.

[3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records diameter in mm under three conditions.

condition	diameter in mm
bright light	2
normal light	4
dim light	7

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to investigate phototropism using seedlings lit from one side.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of coordination and response to an unfamiliar context.

(a) Define homeostasis. [2]

(b) Explain the biological importance of hormones and homeostasis. [3]

(c) Correct this statement: “a learned voluntary action only”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A change in the internal or external environment detected by a receptor.
- 2 B. A cell or sense organ detecting a stimulus.
- 3 C. A muscle or gland that carries out a response.
- 4 D. Maintenance of a stable internal environment.
- 5 A. A rapid automatic response to a stimulus.
- 6 B. Give the complete reflex arc in the correct order.
- 7 C. Accommodation changes lens shape, not the distance between lens and retina.
- 8 D. State stimulus, coordinator, effector and corrective response in homeostasis.
- 9 A. The same auxin distribution can have different effects in shoots and roots.
- 10 C. Cells take up glucose and liver or muscle convert it to glycogen by negative feedback.

Solutions · Structured questions

11. (a) a change in the internal or external environment detected by a receptor [2]. (b) a cell or sense organ detecting a stimulus [2]. (c) a muscle or gland that carries out a response; not a sensory neurone [2]. (d) Credit any two accurate features from the guide: Receptors detect stimuli; impulses travel along sensory neurones to the central nervous system, across synapses using neurotransmitters, then along motor neurones to effectors. Reflexes are rapid automatic responses that protect the body. [2].

12. (a) Insulin secretion increases and glucose falls toward normal [2]. (b) Cells take up glucose and liver or muscle convert it to glycogen by negative feedback [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $7 - 2 = 5$ [2]. (b) The values change from 2 to 7, with the highest value 7. [2]. (c) Radial and circular iris muscles alter light entry to protect the retina and aid vision [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Direction of light [1]. (b) Angle of shoot curvature [1]. (c) Same species, age, water, temperature and time [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Rotate or randomise positions and use several seedlings [1].

15. (a) maintenance of a stable internal environment [2]. (b) Award linked points from: Hormones are chemical messengers transported in blood. Insulin lowers blood glucose by increasing uptake and glycogen formation; glucagon raises it by promoting glycogen breakdown. Negative feedback reverses deviations from a set range. Adrenaline prepares the body for action. [3]. (c) Correct statement: reflex means a rapid automatic response to a stimulus [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.